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The Reliability of HF Propagation Links Over Canada During Absorption Events

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Key Points:

- The reliability of a dedicated HF link between Ottawa and Alert Canada is examined to assess the effect of local auroral absorption regions
- Radio wave absorption measured at 30 MHz at locations in Canada are correlated to instances of signal loss at several lower HF frequencies
- The likelihood of signal reception depends more on the scale size of the precipitation region than the HF frequency used

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Abstract Natural Resources Canada monitors space weather impacts to high frequency radio wave propagation using several High Frequency transmitter/receiver links across Canada. HF data received in Alert from a transmitter in Ottawa are examined in this study. Data for this HF link from 2013 to 2019 at 6 different frequencies between 5 and 14 MHz have been used. For a propagation link between two sites to be viable, two criteria must be met: the overall ionospheric electron density between them must be sufficient to allow the wave to reflect back to the ground and the D-region electron density must be low enough that minimal absorption is experienced by the wave along the path. Both these criteria are impacted by space weather phenomenon affecting ionization. During this period, the University of Calgary was also operating several 30-MHz riometers across Canada. This study examines the relationship between the likelihood of HF dropouts along this link and the absorption measured by the riometer network. Overall, when the F-region density is high enough to support propagation links, losses of the signal at Alert tend to correspond to higher absorption measurements. However, since the absorption measurements are not exactly co-located with the predicted HF link path, measurements of significant riometer absorption (>0.5 dB) corresponded to only ∼30% of signals being lost at all considered frequencies. Overall, the results of this study provide a deeper understanding of the connection between absorption measured by a vertically observing 30 MHz riometer and the corresponding effects on lower frequency oblique HF links.

Plain Language Summary Natural Resources Canada uses long distance radio links across Canada to monitor the impacts of space weather events. This study examines radio data received in Alert from a transmitter in Ottawa. For a radio wave to propagate along a desired path in the ionosphere, the plasma characteristics must meet certain criteria. The peak plasma density must be high enough to cause bending of the wave back down to the ground. Conversely, the plasma density at lower altitudes must be low to reduce absorption of the waves. Separately from the dedicated radio link, the University of Calgary also operates several passive receiving instruments which measure absorption of radio waves called riometers. This study examines the relationship between the likelihood of dropouts along the dedicated link and the absorption measured by the riometers. At times when the ionosphere is expected to support these radio propagation links, losses of the signal tend to correspond to higher absorption measurements. However, since the riometers are not exactly co-located with the radio link, the correlation between absorption and signal loss was relatively weak. Overall, the results of this study improve understanding of how riometer measurements relate to the reliability of Over-the-Horizon radio links.

1. Introduction

The use of High Frequency (HF) radio for communication and defense is an important aspect of our modern technological life (e.g., Fiori et al., 2022; Goodman, 1992; Thayaparan & Themens, 2025; Thayaparan et al., 2018). In particular, in locations with sparse populations and little suitable infrastructure, such as the vast Canadian northern region, very high frequency or ultra high frequency systems which require consistent line-of-sight between transmitters and receivers, are not always practical. Fortunately, at HF, the ionosphere can act as a reflector of radio waves and allow for Over the Horizon (OTH) propagation. Canada is investing heavily into OTH radar systems for defense (Thayaparan et al., 2018). Further, the aviation industry recognizes space weather as an impact to HF communications (e.g., Fiori et al., 2022; ICAO, 2023a). Also, scientific HF radio instruments such as the Super Dual Auroral radar Network (SuperDARN) (Chisham et al., 2007; Greenwald et al., 1995) use OTH principles to remote sense vast regions of the ionosphere and monitor space weather. However, these

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systems can have reduced usability during geomagnetically disturbed periods due to absorption of their probing waves (Bland et al., 2018).

One drawback of using HF waves for reliable communication is the variable nature of the ionosphere which causes variation in the usability of different HF frequencies. If the density of the ionospheric plasma along the radio wave propagation path is not sufficiently high, the waves will not be reflected down and instead escape into space. This results in a maximum usable frequency (MUF) for OTH radio systems that depends on factors such as time of day, location, season, solar activity, etc. Conversely, if the radio frequency is too low, collisions between the charged particles (which essentially carry the signal in a plasma) and neutral particles can cause the wave energy to be lost or absorbed by the medium. As a counterpart to the MUF, the absorption of lower frequency waves results in a lowest usable frequency (LUF) for a given set of ionospheric conditions. Therefore, in general, there is a varying band of frequencies which are usable for OTH propagation for a given time and location (e.g., Athieno & Jayachandran, 2016; Milan et al., 1998; Thayaparan et al., 2018). In practical applications, radio operators desire to know in near-real time what frequencies are ideal for any given set of conditions. Given this and the importance of HF technologies in general, there is considerable effort being made to accurately model and now-cast ionospheric conditions using a wide variety of instruments (e.g., Fiori, Reiter, et al., 2023; Reid et al., 2023).

In general, there are different events that may affect the reliability of HF OTH signals due to either increasing the LUF or decreasing the MUF. In particular, shortwave fadeout, auroral absorption, polar cap absorption, and MUF depression have been identified as the primary impacts to HF radio communications (Fiori et al., 2022). Shortwave fadeouts are caused by solar X-ray flares and result in absorption on the dayside which increases with solar zenith angle. Polar cap absorption (PCA) is caused by solar energetic protons which penetrate the D-region at high latitudes. PCA events can be long-lasting and cause HF communication interruptions at high latitudes for days at a time. Finally, auroral absorption is caused by precipitating high energy magnetospheric electrons at auroral latitudes. As will be discussed, this study uses auroral riometer measurements and a sub-auroral to polar cap HF link, so auroral absorption events are the primary focus. However, it is expected that PCA events may also be of importance. Finally, it can be noted that HF links may be disrupted by MUF depressions caused by a lowering of the F-region electron density after the onset of a geomagnetic storm. These F-region depletions are not considered in this study however, as the focus is on the relationship between riometer absorption measurements and HF communication links only.

Ionospheric radio instruments such as ionosondes, coherent and incoherent scatter radars, and riometers are deployed throughout the Canadian sub-auroral, auroral, and polar regions. However, due to its vast size, it is impossible to sample the entire region at high spatial resolution. Therefore, deployment strategies for ionospheric instruments and how they can be best utilized to provide high spatially and temporally resolved measurements of current ionospheric conditions is an important area of study. Various empirical and theoretical models have been developed to provide the overall morphology of the ionosphere such as the International Reference Ionosphere (Bilitza et al., 2022), the Empirical Canadian High Arctic Ionospheric Model (Themens et al., 2017), and the Assimilated Canadian High Arctic Ionospheric Model (Reid et al., 2023) which can be used to assess the overall usability of HF links. This current study examines the relationship between absorption of vertically incident cosmic radio signals at 30-MHz and the efficiency of HF links at lower frequencies. This study builds on preliminary work done by Ghaly (2024) and will allow for better application of existing riometer measurements to space weather modeling efforts and radio propagation predictions.

As stated above, the general behavior of OTH HF waves propagating in the high-latitude ionosphere is quite well understood. Higher frequencies are expected to be received more often during the day when the ionospheric densities are higher and lower frequencies are expected at night when the D-region densities are lower and there is less overall absorption (Davies, 1990). This study is interested in the relationship between sporadic and relatively short-lived absorption events and HF link quality. The goal is to provide more insight for the observations made by the sparsely distributed riometer network in Canada and how they can best be applied to aid in the operations of HF OTH communications and radar systems in near-real time. Further, it is hoped that this study can inform deployment plans (numbers, locations, types, etc.) for new riometer instruments to improve space weather monitoring in Canada.

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An indication of the relationship between absorption measured vertically at 30-MHz and the corresponding effect on an OTH signal at much lower frequency can be made by examining the Appleton-Hartree equation (e.g., Appleton & Builder, 1933). The full Appleton-Hartree equation is given by:

$$n^2 = 1 - \frac{X}{1 - iZ - \frac{Y_T^2}{2(1-X-iZ)} \pm \sqrt{\frac{Y_T^4}{4(1-X-iZ)^2} + Y_L^2}} \quad (1)$$

Where n is the complex refractive index, X is the squared ratio of the electron plasma frequency to wave frequency, Y_T and Y_L are the ratios of the transverse and longitudinal components of the electron cyclotron frequency to the wave frequency, and Z is the ratio of the electron-neutral collision frequency to the wave frequency. The imaginary portion of the complex Appleton-Hartree equation describes decay/power loss of the wave as it travels through the medium. The imaginary portion of Equation 1, along with the wave equation, can be used to determine the absorption of a wave as it propagates through a plasma by:

$$A(s) = 4.6 \times 10^{-5} \int \frac{\upsilon N}{\omega^2 + \upsilon^2} ds \ \ [dB] \quad (2)$$

Where A is the absorption in dB, υ is the electron-neutral collision frequency, ω is the angular wave frequency, and N is the electron density. All parameters in Equation 2 use SI units.

In general, absorption of propagating HF waves occurs in the lower parts of the ionosphere where there are both significant amounts of free electrons and neutral particles (which cause collisions with the electrons before they can re-radiate the wave energy). Further, the absorption experienced by an HF wave depends on its total path length and is roughly proportional to the inverse of frequency squared (note that, typically υ is much less than the wave frequency, so the absorption in Equation 2 is roughly proportional to $1/f^2$). Since an OTH wave passes through the D-region of the ionosphere at least twice, at a relatively low elevation angle, and is typically <~15 MHz, it will experience significantly more absorption than the cosmic signal observed by the vertically pointing 30-MHz riometers. In fact, as an example, a 0.5-dB measurement of absorption from a 30-MHz vertically pointing riometer should correspond to ~50 dB loss for a typical OTH wave at 10 MHz (assuming the elevation angle is ~10°). Therefore, any absorption measurement that is distinguishable from the background noise level of the riometer would be expected to almost always correspond to nearly total signal loss on a typical HF link between two sites if the same absorption region was encountered, which is not always true. The main goal of this study is to develop a better understanding of how a measurement of absorption made by a riometer can inform the behavior of oblique HF propagation which may be offset by several hundred km or more.

2. Instrumentation

2.1. NRCan HF Transmitter/Receivers

Natural Resources Canada (NRCan) operates a network of HF transmitters and receivers across Canada (Cameron et al., 2021; Danskin et al., 2018; Fiori, Rogers, et al., 2023). One of the purposes of this network is to examine and monitor the effects of ionospheric absorption on HF communications in the auroral and polar regions. During routine operations, the transmitters typically operate at 6 fixed HF frequencies (i.e., 5.38, 6.93, 8.10, 10.42, 11.11, and 14.36 MHz). For a given transmitter/receiver pair, each frequency is transmitted as a continuous wave followed by a Barker coded sequence for a few seconds each. For a given transmission, if the receiver successfully decodes the Barker code, it records the signal power, angle-of-arrival (elevation and azimuth), Doppler shift, and propagation time for that sequence. Each frequency is transmitted at a set known schedule with 4, 8, or 24 transmission sequences every hour. Reception data from the Ottawa to Alert link during the years 2013, 2014, 2015, 2017, and 2019 have been made available for the scientific community (Cameron, 2020) and are used in this current study.

The locations of the NRCan HF sites used in this study are shown in Figure 1. Note that the path from Ottawa to Alert is over 3,000 km in a roughly south-to-north direction and passes from sub-auroral, to auroral, to polar latitudes. HF propagation modeling and analysis of the received propagation delays suggests that at least 2 hops are required for an OTH wave to propagate from Ottawa to Alert (Cameron et al., 2021). In other words, the radio

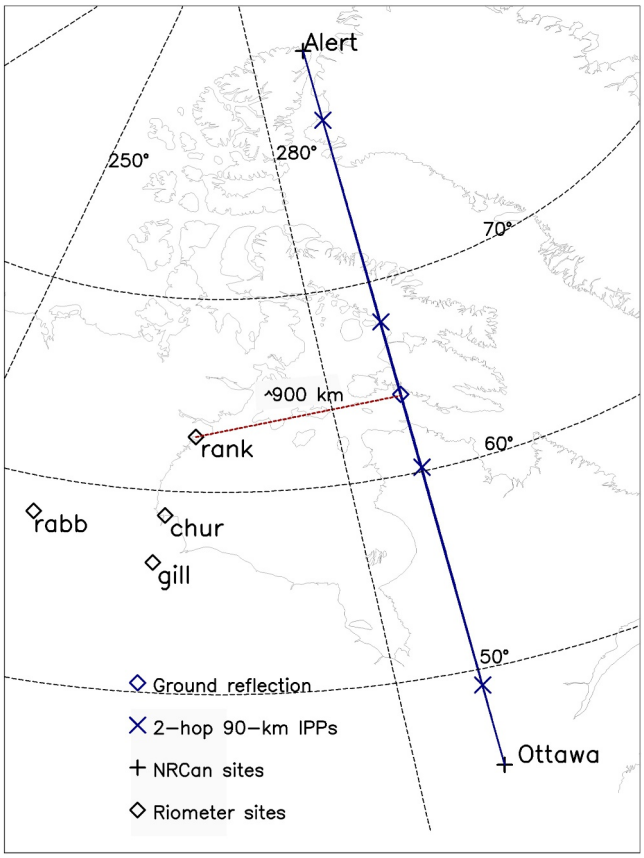


Figure 1. The locations of the GO-Canada 30-MHz riometers used in this study and the Ottawa and Alert NRCan transmitter/receiver pair. The blue line shows the predicted HF path from the Ottawa transmitter to the Alert receiver. The blue *x*-symbols along this path represent the approximate location of the D-region ionospheric pierce points for 2-hop propagation. The blue diamond is the midpoint of the propagation path, where the wave is expected to reflect off the ground the first time. The red dashed line shows the distance between Rankin (the closest riometer) and the midpoint of these pierce points.

wave path must reflect from the ground at least once between the two sites (and therefore pass two additional times through the absorbing D-layer). Assuming a somewhat laterally uniform ionosphere, these two middle D-region ionospheric pierce points (which are closest to the auroral zone and would be most likely to be affected by absorption) would be near 60° and 68° geographic latitude. It may be noted that in this study, the D-region ionospheric pierce points are all modeled to be at 90-km altitude. All four ionospheric pierce points are marked by the blue crosses in Figure 1. The midpoint between the transmitter and receiver sites (where the wave is expected to reflect from the ground for 2-hop propagation) is shown as the blue diamond. The red dashed line from Rankin Inlet to the propagation path illustrates the distance between the HF link and the riometer sites (~900 km). The pierce points in Figure 1 show where the HF waves would cross 90-km altitude and were calculated by assuming straight-line propagation below 90 km at 10° elevation (the typical elevation angles measured at Alert). More realistically, due to solar zenith angle effects, there is typically a latitudinal gradient in ionospheric electron density, so the actual pierce points are probably slightly closer to Ottawa. However, since this study is considering several years of absorption and HF reception data, the ionospheric profile will be variable, so only the pierce points assuming a simple laterally uniform profile are shown as a general estimate of their actual locations. Similarly, for 3 and 4 hop propagation, one might expect ground reflections at roughly equal distances along the propagation path between Ottawa and Alert. The exact locations will vary depending on the horizontal F-region density profile and the frequency being used, but in general, it is expected that the path will have passed through the auroral D-region at least twice for all receptions (the initial up and final down D-region passes are most likely in the sub-auroral ionosphere and polar cap, respectively).

2.2. GO-Canada Riometers

The University of Calgary operates a network of 30-MHz wide-beam Geospace Observatory Canada (GO Canada) riometers across Canada (Rostoker et al., 1995; Spanswick et al., 2024). These riometers have been operating mostly continuously for several decades, including the 2013–2019 timeframe of this study. Each riometer observes the cosmic radio background in the vertical direction throughout a given day. For a given sidereal time, this radio source is relatively constant, so any deviations from it can be attributed to absorption of the 30-MHz signal by the ionosphere. In practice, since environmental and hardware conditions at each site can change gradually or suddenly, calibration of the data for a given day and site is required. This baselining or quiet day curve (QDC) subtraction converts the raw signal power measured by a riometer into a measurement of the absorption a 30-MHz wave experiences as it passes vertically through the D-region (NORSTAR, 2014). Note that small deviations in modeling the QDC can result in a <0 dB absorption value. Although non-physical, these values simply represent fluctuations due to signal noise around the QDC when there is essentially no absorption present. These values have been kept in the analysis to prevent biasing the results toward positive absorption values at times when there is no absorption. It can also be noted that the typical baseline subtraction effectively removes any background diurnal trends that may be present in the D-region. Therefore, measurements by the riometers represent anomalous absorption above the expected diurnal background at each time of day. According to the imaginary portion of the Appleton-Hartree equation, the absorption measured at 30 MHz is related to the absorption at any other frequency through an inverse frequency-squared relationship (though analysis of multi-frequency absorption data shows this is not always the case (Fiori et al., 2025)).

The locations of the riometer sites are also shown in Figure 1. Note that all the riometer sites are to the west of the OTT-ALE link. During this period, 12 GO-Canada riometers were typically operating, however this study uses data from only the 4 closest sites to the predicted HF link, Rankin Inlet (rank), Churchill (chur), Gillam (gill), and

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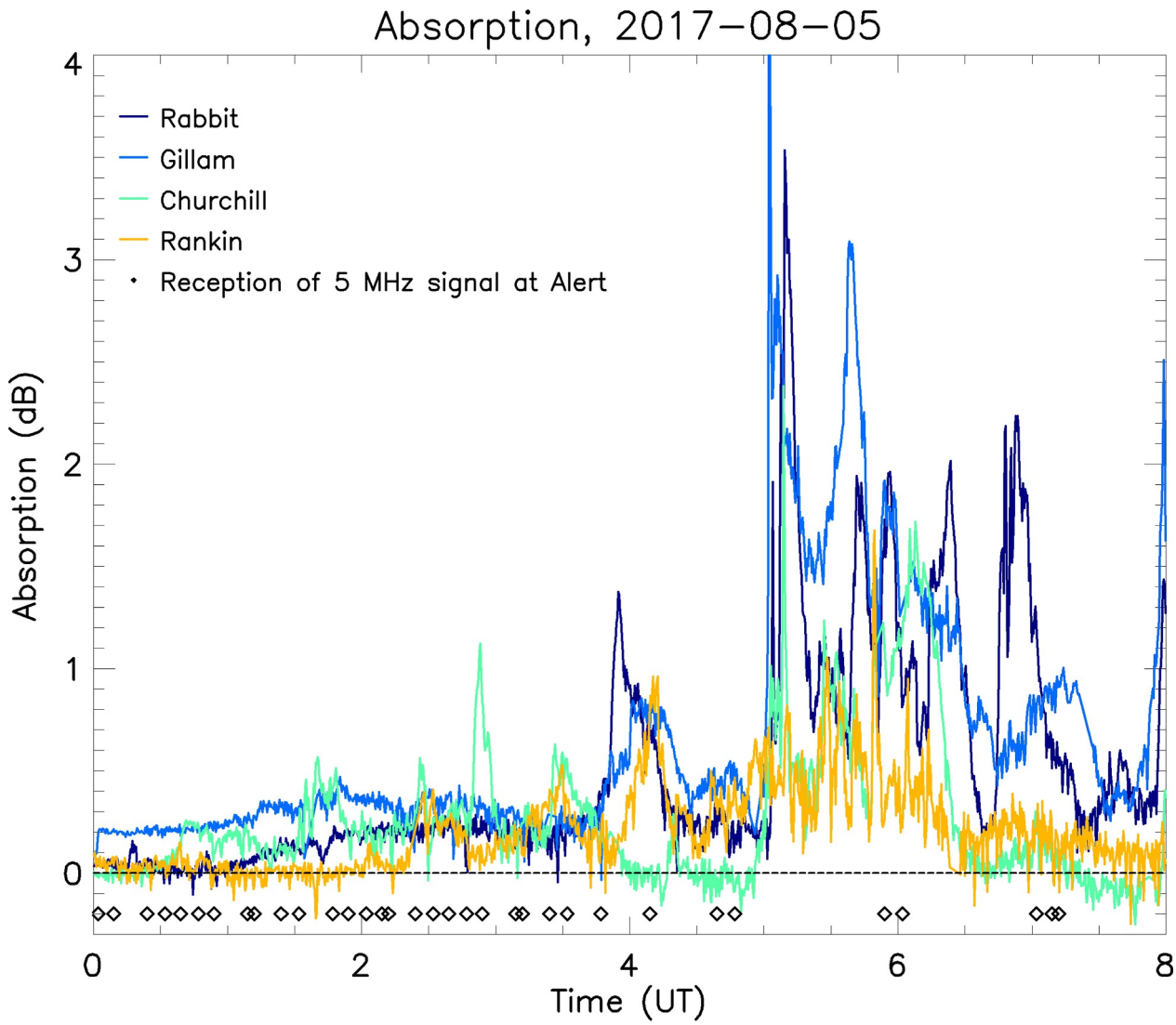


Figure 2. The absorption measured at Rabbit, Gillam, Churchill, and Rankin Inlet on 05 August 2017 from 0 to 8 UT. The black diamonds near the bottom of the plot show the times when the 5-MHz signal from Ottawa was received at Alert.

Rabbit Lake (rabb). The riometers measure absorption at 5-s cadence, however, for this study, 1-min averages are used. It can be noted that the East-West distance from these sites to the assumed HF propagation path is at least ~900 km. Further, as mentioned, the ionospheric pierce points of the HF path may be several hundred km north of the typical auroral riometer sites (Gillam and Churchill). Therefore, overall, considerations must be made as to whether an absorption region (or lack of one) observed above these riometer sites will be representative of the conditions encountered by the HF signal. A typical auroral absorption event measured by these riometers on 05 August 2017 is shown in Figure 2. It can be observed in Figure 2 that a strong auroral absorption event can often have values up to 3–4 dB (or higher) at 30-MHz and last for several minutes to hours. Also shown in Figure 2 as black diamond symbols are the times when the 5-MHz NRCan transmissions from Ottawa were received at Alert during these 8 hr. During this period, the transmission schedule was such that 8 pulses per hour at fixed times were sent. When the absorption recorded at the riometers increased significantly at ~5 UT, the receiver in Alert stopped observing the 5-MHz signal. Although, it can also be noted that later at ~6 and 7 UT, there was still significant absorption at most riometer sites, but some signals did arrive at Alert at this low HF frequency.

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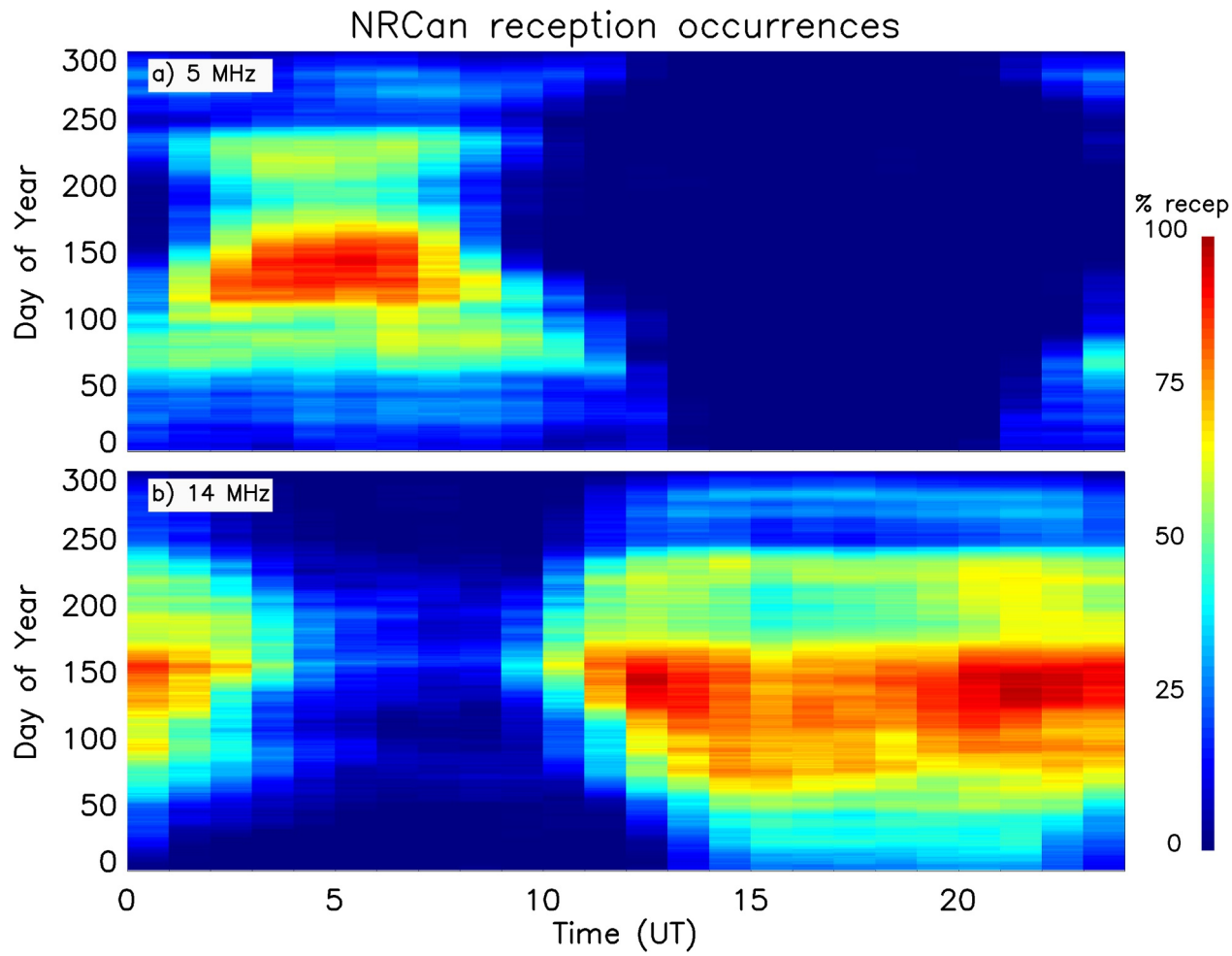


Figure 3. (a) The reception occurrence percentages as a function of universal time and day-of-year for (a) 5 MHz and (b) 14 MHz.

3. Results

3.1. Reception Occurrences

Overall, ionospheric and HF ray path modeling predicts some general trends expected in the reception occurrence of signals at Alert for different propagation frequencies. For example, it is expected that the higher frequencies (e.g., >10 MHz) will be received at Alert more often during the daytime than the nighttime, due to the requirement that the F-region density be sufficiently high to reflect signals. During nighttime, the ionospheric density can decrease below a level that allows propagation from Ottawa to Alert on the higher frequencies. Conversely, the lower frequencies (e.g., the 5 and 6 MHz bands in particular), will almost always be reflected by the ionosphere, even at nighttime. However, the lower frequencies are also significantly more affected by absorption, so during the daytime when the D-region density is higher overall, they are lost in that manner instead. Other factors, such as time of year, solar activity, and location affect the reliability of a given HF link, but in general the diurnal variation with frequency is the most significant for this experiment. It can be noted that Cameron et al. (2021) performed a thorough analysis of the reception occurrence as a function of frequency, time of year, and time of day (e.g., Figure 6 of that study).

Figure 3 shows the reception occurrence of signals at Alert from Ottawa for the 5 and 14 MHz bands for a given time of year and day. In generating Figure 3, it was assumed that the transmitter and receiver were both operating continuously on the days when data was recorded. The exception to this was data from July and August 2016. These dates were excluded because, compared to the same periods in other years, there was almost no received data, and it was therefore assumed that at least one side of the HF link was not operating continuously during this

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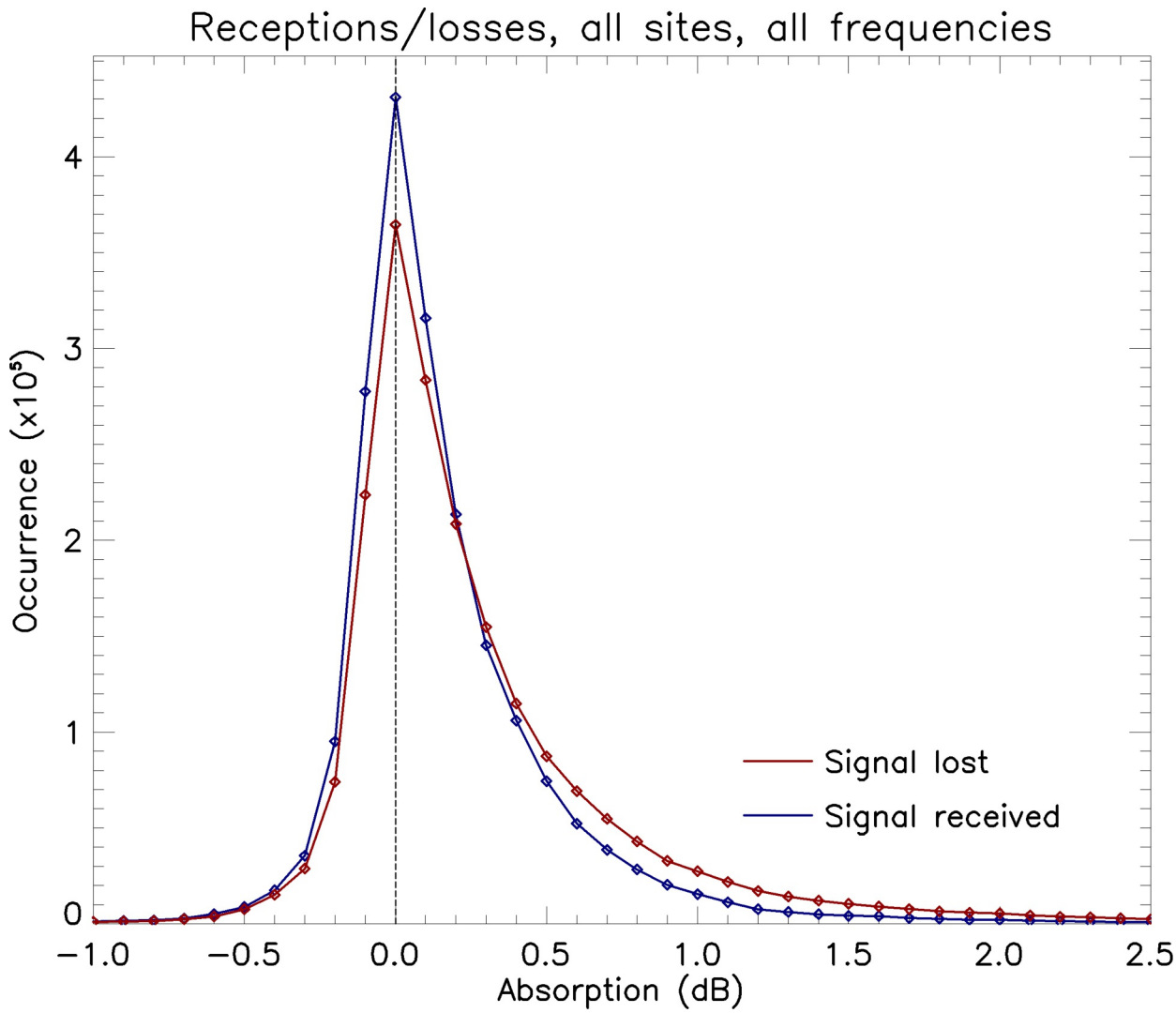


Figure 4. Blue curve: the number of receptions at Alert as a function of absorption level at all sites for all HF frequencies. Red curve: the number of lost signals for all frequencies as a function of absorption level at all sites, normalized to the area under the blue curve.

time. The percentage displayed is then the recorded number of pulses received in each hour divided by the number of pulses transmitted as outlined in Cameron et al. (2021). For reference, along the north-south oriented propagation path, local noon is ∼19UT. As expected, the lower frequency (5 MHz) band shows more receptions at night, while the higher 14-MHz band shows more receptions during the day. The diurnal variation observed in the 5 MHz reception occurrence during summer months, when a significant portion of the Ottawa-Alert HF path is continuously sunlit, is driven by the solar zenith angle dependence on D-region ionization. Though not shown, the intermediate frequencies (6, 8, 10, and 11 MHz) show occurrence behaviors which are in between these two extremes.

3.2. Signal Loss Versus Absorption

All available Ottawa-Alert reception data from 2013 to 2019 were considered. For each hour the transmitter and receiver were operating, between 4 and 24 pulses are expected to be received. Figure 4 shows as the blue curve the total number of pulses received for these time periods for a given absorption level measured at the riometer sites. In Figure 4, results from all HF frequencies are included and the absorption values are the average of all the sites for the minute when the HF pulse was expected to be received at Alert. The red curve in Figure 4 is the same as the blue curve except it shows the number of pulses that were not received. The number of pulses not received for a

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given hour is simply the number of pulses received in that hour subtracted from the total number of pulses transmitted per hour at that frequency (from the set transmission schedule). Further, the red curve has been scaled so that the total area under it is equal to the total area under the blue curve so that a direct visual comparison of the distributions may be made.

Figure 4 shows that when the riometers measure very little to no absorption (e.g., below 0.2 dB), relatively more pulses are received than lost. Conversely, as absorption increases, more signals are lost than received, as expected. However, somewhat surprisingly, at 0.5 dB and above, despite relatively more signals being lost, the HF link is still available a significant amount of the time. As discussed, it is expected from the Appleton-Hartree frequency relation and the oblique nature of OTH propagation that 0.5 dB of absorption measured at 30 MHz would correspond to ∼50 dB signal loss at these lower frequencies. It can be noted that the study by Cameron et al. (2021) examined the statistics of received power at Alert from Ottawa and found the dynamic range of received signal powers spanned about 50 dB at most. So, if a 0.5 dB absorption vertically at 30 MHz corresponds to a 50 dB reduction in power for a 10 MHz oblique OTH wave, then it would be expected that the same absorption region should result in complete signal blackout for all the NRCan operating frequencies except possibly 14 MHz. Therefore, consideration of the distance between the riometer measurements and the HF link is clearly important.

Figure 5 shows the signal loss probability (the number of lost pulses divided by the total number of expected pulses) for different absorption levels. The curve shows the overall average loss occurrences using all HF frequencies and absorption averaged from all riometer sites in increments of 0.1 dB with bars indicating the binomial standard error for each bin. Again overall, as measured absorption increases, more signals are lost. However, for this experimental set-up, there is no threshold absorption value at which all HF signals are lost as may initially be expected. More detailed comparisons between different riometer sites and OTH frequencies will be examined further in the following sections. It should be noted that over the entire data set considered, in total 62% of expected pulses were not received at Alert (likely either due to absorption of the signal or lack of a suitable propagation path). This overall loss probability is shown as the horizontal dashed line in Figure 5 and serves as a baseline representing the expected loss probability independent of absorption. This is analogous to a risk ratio, where the probability of receiving the signal at a given absorption level can be compared directly to the baseline reception probability of 38%. Therefore, for a given absorption level, if more than 62% of the pulses did not arrive at Alert, there were more signals lost than would be expected if there was no absorption dependence. It can be observed that below 0.2 dB the signal loss probability is below the baseline, indicating fewer losses than average. Conversely, as measured absorption increases, the loss probability rises, reaching ∼80%–90% above 2 dB.

3.3. HF Link Frequency Dependence

The relationship between the number of signal receptions or losses and absorption is examined with consideration of the transmitted frequency. Figure 6 is similar to Figure 5, however the loss percentage as a function of absorption is shown for each of the different operating frequencies. Also, in Figure 6, it is the *relative* change in signal loss probability that is shown (not the absolute loss probability). For each frequency, the difference between the loss probability at a given absorption level $p(A, f)$ and the overall baseline loss probability for that frequency $P_{\text{baseline}}(f)$ is normalized by the maximum possible increase in loss probability. The relative change in signal loss probability $P_{\text{rel}}(A, f)$ is then given by Equation 3:

$$P_{\text{rel}} = 100\,\frac{P - P_{\text{baseline}}}{100 - P_{\text{baseline}}} \tag{3}$$

A value of 100% would therefore indicate that all signals that would normally be received were lost. Therefore, values in Figure 5 below the horizontal dashed line would give negative values if presented in the same manner as Figure 6, while values above the overall loss probability would give positive values. For each frequency, the relative change in loss probability increases as the measured absorption increases. Conversely, when there is very little or no absorption in the region, pulses sent from Ottawa to Alert are observed more often. Further, it can be observed that, possibly somewhat surprisingly, the frequency of the HF link seems to have little effect on the relative loss percentage curves. The exception to this is the 14 MHz results which do tend to show less signal loss compared to the other frequencies as expected from Equation 2. Overall, this behavior does not immediately follow from the wave propagation theory since higher frequencies should have less absorption. However, as

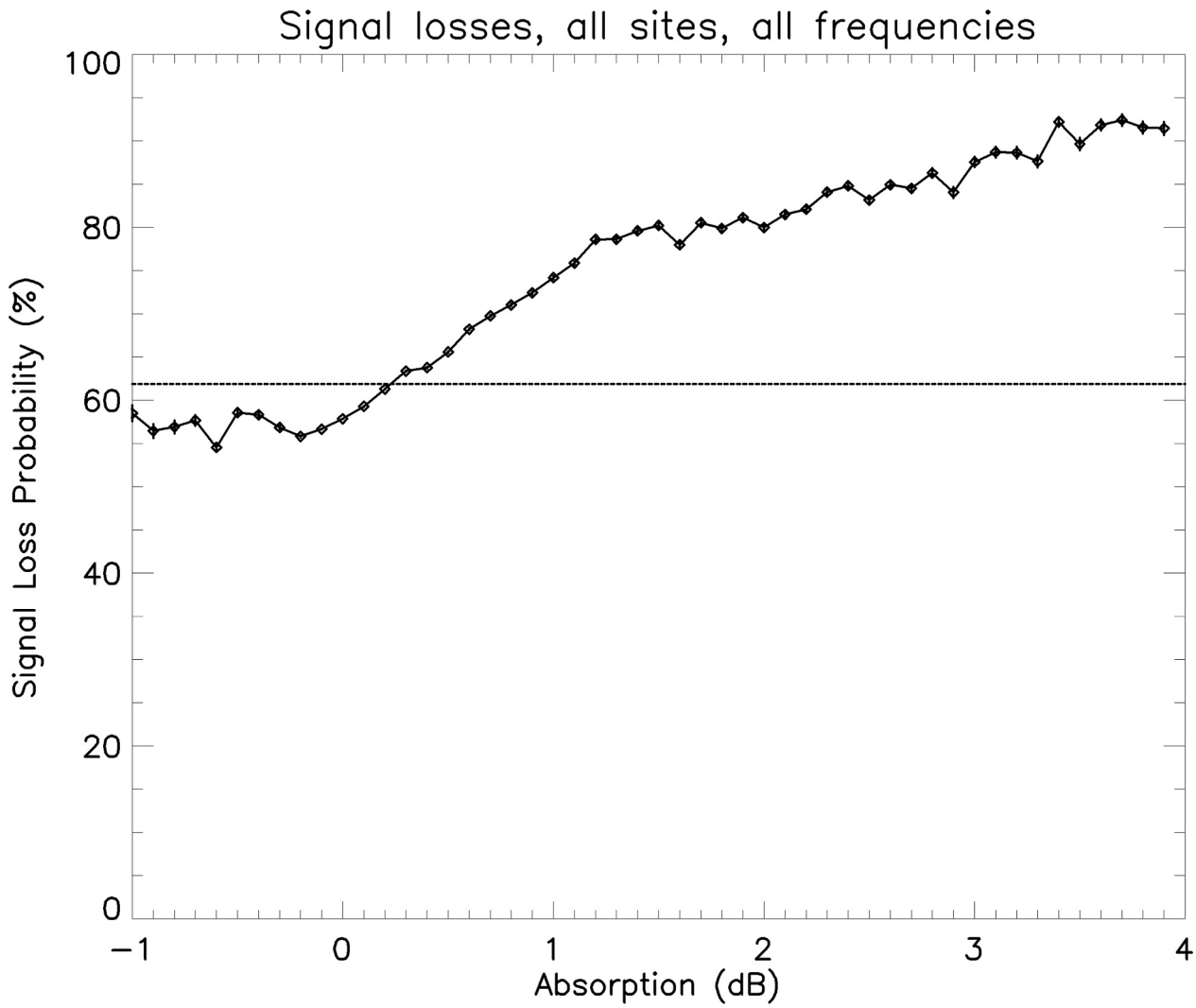


Figure 5. Signal loss probability as a function of measured absorption. For each absorption bin, the loss probability is the percentage of expected pulses not received at Alert. Error bars show the binomial standard error for each bin. The black dashed line shows the overall signal loss probability for all absorption levels for the times considered in this study.

discussed, the absorption measured vertically at 30 MHz is orders of magnitude less than the absorption of an OTH wave at these lower frequencies. Therefore, the driving factor that determines if the HF link is lost or not will be whether the absorption region observed above the riometer sites extend spatially to the D-region pierce points of the HF link and not on the frequency of that HF wave. Gillies et al. (2024) showed that the scale size of the absorption regions measured by this riometer network are highly variable. Typically, the regions are smaller than the distance between the closest sites in this study and the HF link, however, often the region is highly extended (in particular, in the East-West/zonal direction). A deeper examination of absorption region scale sizes with respect to the results of this study is discussed in Section 4.

3.4. Riometer Location Dependence

As has been shown in Figures 4–6, the HF link between Ottawa and Alert is lost more often when there is increased absorption at least one of the four sites considered in this study. Figure 6 showed that there is little difference in the signal loss probability for different frequencies. The next relationship to examine is the dependence of site location on reception occurrence. This is essentially examining the effect of measuring absorption at different distances from the HF link path and D-region pierce points.

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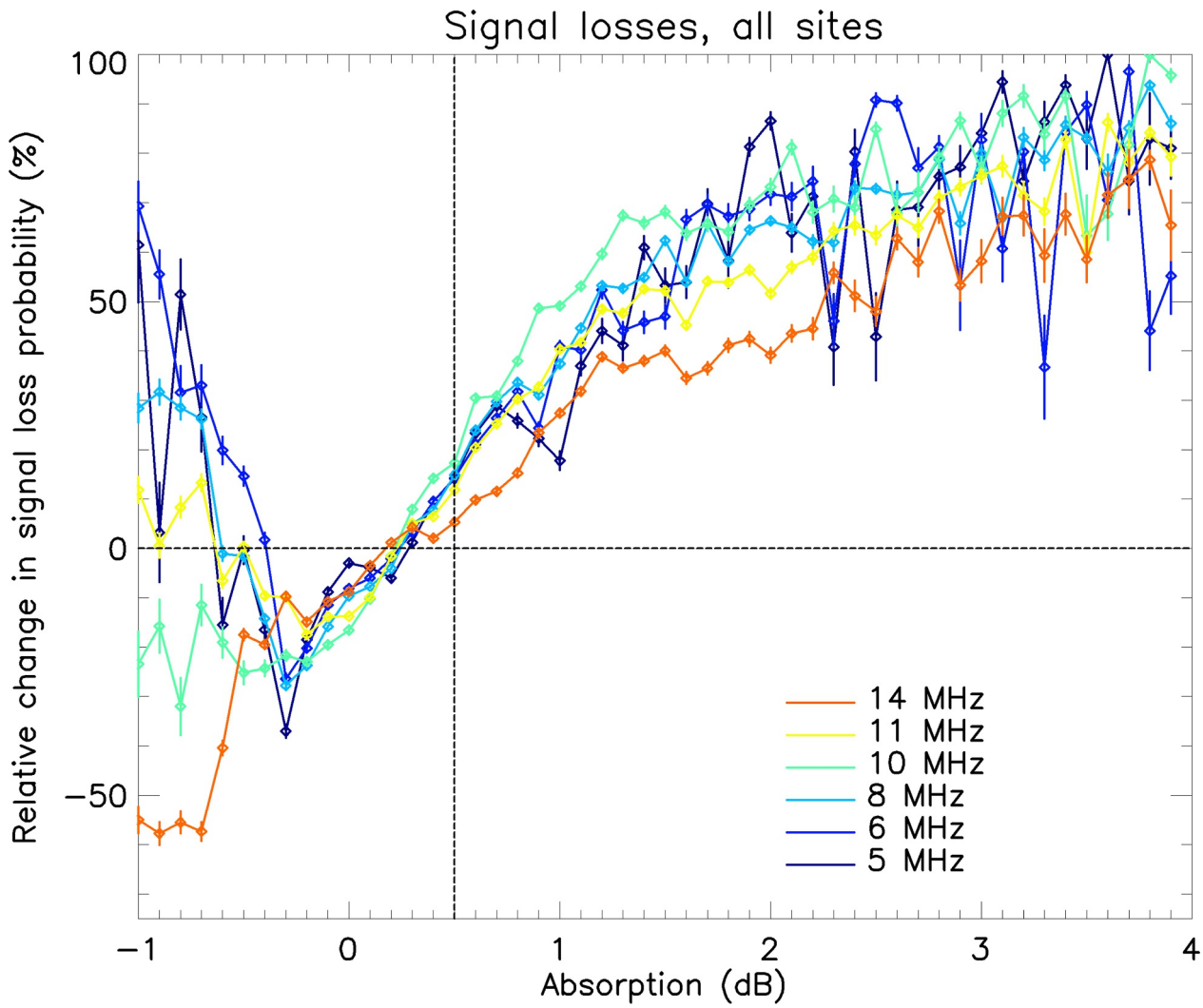


Figure 6. The relative change in HF signal loss probability as a function of measured absorption for different frequencies. For each curve, the loss probability is normalized by the overall signal loss probability for that frequency, such that positive values indicate more signal losses than expected. The absorption values are averages of all included sites.

Figure 7 is in the same format as Figure 6 but presents the relative change in signal loss probability as a function of measured absorption for different riometer sites. As shown in Figure 1, Rankin Inlet is the closest riometer site to the predicted HF link path while Rabbit Lake is the furthest. Churchill and Gillam are close to the same distance from the path (since the exact locations of the D-region pierce points will vary, it is difficult to say which is closer). Overall, the absorption measured at Rankin Inlet corresponds most closely with more instances of signal loss than the other sites. This may be expected as it is both closest to the expected middle ionospheric pierce points for two-hop propagation and it is the furthest outside the typical auroral zone, so any absorption measured there indicates a large event. For example, 0.5–1 dB of absorption at Rankin Inlet is much more significant in terms of the number of signals lost than a similar measurement at Rabbit, Gillam, or Churchill and one would expect a more pronounced effect on nearby HF links.

In general, the trend is that absorption measured at the closest site to the HF link/D-region pierce points, Rankin Inlet, tends to have a stronger correlation with the HF dropouts. The other three sites are farther away and show less predictive capability on HF link status. A 50% increase in signal loss probability occurs when Rankin Inlet observes ∼1 dB or higher. Moving away from the HF link, a 50% increase in signal loss occurs when the auroral sites (Gillam, Churchill, and Rabbit) observe 1–2 dB or higher absorption. These values provide information on how applicable riometer measurements may be at varying distance from communication links of interest.

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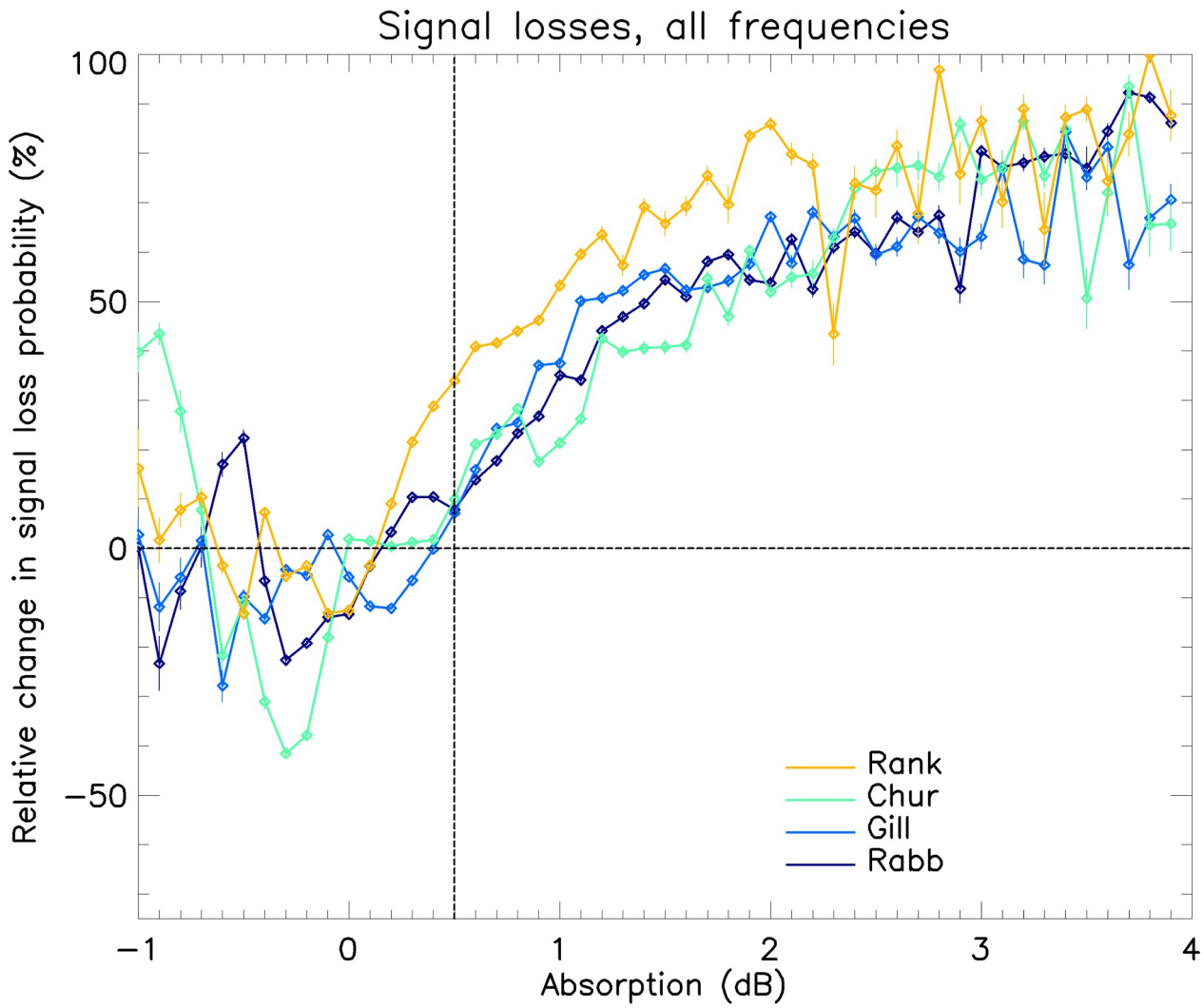


Figure 7. The same as Figure 6, except for different riometer sites and loss probabilities averaged over all HF frequencies.

4. Discussion

The effect of auroral absorption, as characterized by absorption measured at 30 MHz, on the reliability of HF OTH links for propagating signals in the ~5–15 MHz range has been examined. Overall, it has been found that as measured absorption increases, there are fewer HF signals recorded at the receiver site as expected. However, there is not a sharp cut-off absorption level, above which no reception occurs, nor does there appear to be a strong dependence on the HF frequency that is used (at least from 5 to 11 MHz). The weighted average of the additional signal loss probability above 0.5 dB was calculated to range from 26% to 32% for most frequencies. The 14 MHz results do tend to show fewer signal losses compared to the other frequencies for a given absorption level (the weighted average above 0.5 dB was only 19%). This behavior is expected based on the frequency dependence in the Appleton-Hartree equation. Also, it may be argued based on Figure 7 that the 10 MHz results tend to show more losses than the other frequencies (a weighted average of 40%), but the possible reasons for this behavior are uncertain. However, despite some variation, overall, there does not seem to be a clear frequency dependence explainable by the $1/f^2$ dependence only.

Attempts have been made to establish a cut-off for the purpose of space weather alerting services. For example, based on an evaluation of SuperDARN data, Fiori, Rogers, et al. (2023) determined that signals in the 5–15 MHz range are at risk when the absorption at 30 MHz exceeds 0.5 dB, but suggests a 1.0 dB threshold to indicate more spatially widespread impacts to the full frequency range. It should be noted that these cut-offs were based on

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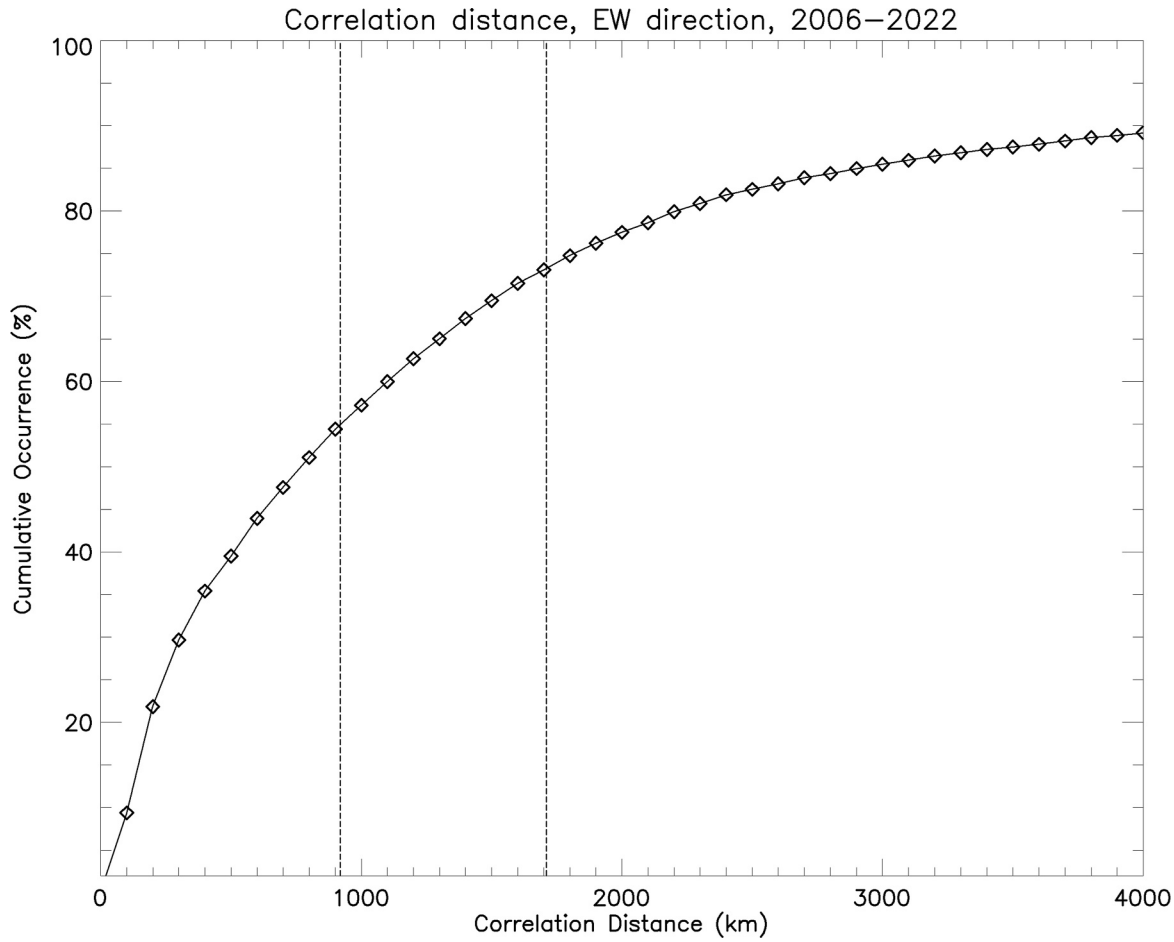


Figure 8. The cumulative occurrence of auroral absorption events with different correlation distances/scale sizes using the entire network of GO-Canada riometers and data from 2006 to 2022. The left (right) vertical dashed line shows the approximate distance from Rankin Inlet (Rabbit Lake) to the D-region pierce points.

riometers located at corresponding SuperDARN transmission points, and not D-region pierce points. To ensure impacts across a broad band of propagation paths and frequencies, the space weather service operated on behalf of the International Civil Aviation Organization suggests thresholds of 2 and 5 dB to characterize moderate and severe degradation to HF communications (ICAO, 2023b).

The lack of a sharp cut-off determined in this study is initially counter-intuitive, as the Appleton-Hartree formula predicts that a measurement of absorption vertically at 30-MHz should correspond to orders of magnitude more absorption of a 2 or more hop HF signal which passes through the D-region multiple times at low angles. However, HF signals were frequently received when absorption was measured at the riometer sites. As discussed in Section 1, the Appleton-Hartree equation predicts that the electron density at the D-region pierce points must have been relatively low for those signals to have been received. Therefore, since the riometers measured significant absorption, which was apparently not encountered by the HF wave along its path, the absorption region observed above the riometer sites must not have extended to the D-region pierce points of the HF link. Given that the riometer sites are separated from the pierce points by 900–1,700 km, this implies that the absorption regions responsible for the riometer measurements were smaller than this separation distance in those cases. Further evidence for this behavior is the fact that sites closer to the HF link show a higher correlation with signal losses than sites further from the link. Conversely, it can be assumed that there are instances when signal is lost at Alert, but no absorption is measured at the offset riometer sites, however these situations are not considered here.

Gillies et al. (2024) performed a statistical examination of over 17 years of GO-Canada riometer data to estimate the typical scale size of observed auroral absorption regions over Canada. The distribution of these scale sizes based on correlations between each riometer site was found to be quite variable. For each absorption event

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examined in the Gillies et al. (2024) study, the decrease in correlation between sites as a function of distance between them was examined. An exponential function was fit to each of these correlation-distance curves and a proxy for the scale size of an absorption region was determined from the decay term of these fits. Often the estimated size of the absorption regions was found to be even larger than the entire GO-Canada array. Since the scale size proxies were determined from fits to the correlation-distance function, events that had high correlation across the entire riometer array could result in estimated absorption region sizes that were larger than the array. However, typically, the proxy for the scale size of the absorption regions were smaller than ∼1,000 km in the magnetic East-West direction. Therefore, it is not unexpected that there is some (but far from perfect) correlation between the observation of absorption at Gillam, for example, and the loss of signal along the HF OTT-ALE path.

Figure 8 shows the cumulative occurrence of auroral absorption region scale sizes in the magnetic East-West direction observed by the GO Canada array from 2006 to 2022. The value shown on the vertical axis is then the percentage of absorption events which had scale sizes smaller than a given correlation distance on the horizontal axis. The procedure used by Gillies et al. (2024) was employed to generate these scale size estimates. Each absorption event with measurements from at least three riometers in this 17-year period was used to estimate the scale size by correlating the absorption data between pairs of sites and examining the decrease in correlation as distance between site locations increased. The vertical dashed lines indicate the rough distance from the Ottawa-Alert HF link D-region pierce points to the closest (Rankin Inlet, ∼920 km) and furthest (Rabbit Lake, ∼1,710 km) riometers used in this study. For a given measured absorption event to affect the HF link, the size of the region would have to be at least this large. It can be observed that most (∼55%–75%) of the events measured by the riometers have scale sizes which are smaller than this distance. Therefore, at most, only ∼25%–45% of the events measured by the riometer chain used in this study would be expected to affect the HF link. A 0.5 dB threshold for identifying an absorption event was used in generating the scale sizes shown in Figure 8. For absorption data above 0.5 dB in Figure 7, a weighted average of the additional signal loss probability ranged from 24% to 28% for the three furthest riometer sites and was 41% for Rankin Inlet, the closest site. This is in general agreement with the findings of the Gillies et al. (2024) study which showed that absorption regions with scale sizes exceeding the riometer-to-link distances only occurred 25%–45% of the time.

5. Summary and Conclusions

Space weather impacts on the reliability of OTH radio links in the 5–15 MHz range over Canada have been examined by comparing signal reception and measurements of vertical absorption made at 30 MHz. Data from a dedicated HF radio link between Ottawa and Alert (roughly south-to-north, crossing from sub-auroral to polar latitudes) from 2013 to 2019 is used in conjunction with riometer measurements of vertical ionospheric absorption which are offset by varying distances from the predicted D-region pierce points of the HF wave path. Overall, it is observed that as measured absorption increases, the HF signals which would propagate from Ottawa to Alert under normal conditions tend to be lost more often. However, even at 30-MHz vertical absorption values above 0.5 dB, signals are only lost due to absorption ∼30% of the time, when theory expects an oblique HF wave at ∼10 MHz would be completely absorbed if it is assumed that the D-region along the oblique path has a similar electron density. Despite initial assumptions that lower frequencies would experience more signal losses than higher, this behavior was not observed with the HF frequencies used in this experiment (with the possible exception of the highest frequency–14 MHz). The parameter that more closely predicted the HF signal loss was the proximity of the riometer site to the HF link path. The absorption to signal loss correlation was strongest for the Rankin Inlet riometer which is closest to the ionospheric pierce points. The increase in signal loss probability as measured absorption increased was in general agreement with the estimates of the East-West extent of the absorption region scale sizes. This suggests that riometer measurements close to operational HF links are highly desired as the predictive ability of these measurements drops off significantly if they are more than a few hundred km away. These findings will hopefully aid in future space weather monitoring strategies for deployment of riometer sites across large landmasses such as the Canadian north and allow better application of the existing sparse absorption measurements for radio operators.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

Riometer data used in this paper can be found in the GO-RIO data archive (Spanswick et al., 2024), which is maintained by the University of Calgary. Both raw and baselined GO-Canada riometer data can be found in this archive using the following URL, organized by date: Index of/sort_by_project/GO-Canada/GO-Rio/txt/. NRCan HF transmitter network data is available online (Cameron, 2020).

Acknowledgments
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